VMware goes mobile
VMware Inc., the leader in desktop virtualisation, recently announced its Mobile Virtualisation Platform (MVP) for mobile phones

Growth ahead
A recent Gartner report predicts that by 2012, 50 per cent of new smartphones will use mobile virtualisation

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Virtualisation Goes Mobile

We take a look at mobile virtualisation and what it means for the smartphone market

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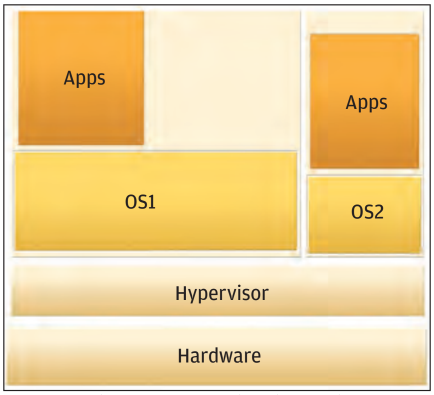
Have you ever downloaded an application to your smartphone, only to find that it is not compatible with the operating system installed on your phone? Have you ever bought a new cell phone, only to realise that all your carefully backed up applications are incompatible with your new phone? The latest smartphones come installed with a variety of operating systems, all of them incompatible with each other. To top it off, none of these operating systems has a clear advantage. This confusion forces you to make trade-offs, trying to pick one operating system over the others. Your decision will usually depend on the application ecosystem, and its overlap with your requirements. As a result, no such decision will be perfect, because sooner or later you will run across an application which you want but cannot have. Mobile virtualisation is a potential solution to such issues. In fact, enterprise research firm Gartner has predicted that by the year 2012, more than 50 per cent of all new smart phones will contain some kind of virtualisation technology built into the handset by the manufacturer.

Over the last few years, virtualisation has developed from a niche technology to one that is being increasingly used to run servers and data centres more efficiently than ever before. Most servers used for different applications usually run at 20 to 40 per cent of their maximum capacity. This is required since we need to provide a buffer for short periods when there may be extremely high load on the server. The rest of the time, the server stays idle with low utilisation. With multiple servers installed in a data centre, you can see that the average utilisation will be low while the power consumption and cooling expenses will be much higher. Though virtualisation was initially implemented by IBM for increasing the utilisation of its mainframe systems, it was only recently that people started using it on normal servers as well. So how exactly does virtualisation help? Here's how...

The core of virtualisation technology is a piece of software called a 'hypervisor'. This hypervisor runs on top of the system hardware in the form of a thin layer or interface between the hardware and the operating system. The advantage of the hypervisor is its ability to support multiple operating systems running in parallel on top of the hypervisor. Each operating system behaves as a separate machine, and has access to the hardware of the base machine through the hypervisor software. Each operating system is known as a virtual machine. Clearly, the advantage of this approach is that multiple virtual machines can be running at the same time on a single piece of hardware. The resources of the base machine are shared among the virtual machines by the hypervisor. The resource allocation can be changed dynamically according to demand to give more resources to virtual machines that are under a higher load. Virtual machines can even be moved from high utilisation machines to those with lower utilisation without having to shutdown the machine. Virtualisation also simplifies backup and disaster recovery processes since the virtual machine can be copied to a backup location and recovered in the same way from the backup. As a result of implementing virtualisation, the average utilisation of the base server hardware is much higher. This in turn results in cost savings in terms of hardware, power and cooling expenses. Virtualisation has been used for testing multiple operating systems as well as running web servers which have low average utilisation among other uses. Mobile virtualisation is the latest form of virtualisation to emerge in the IT industry.



Two operating systems running side by side on a single server

Mobile virtualisation is similar to server virtualisation, except that here we are running multiple mobile operating systems on top of a small software hypervisor. Due to the limitations of these mobile platforms, available memory is much lesser, requiring slim hypervisors which will not use large amounts of memory. The basic concept remains the same with some slight changes. For example,